

DUAL NATURE OF RADIATION AND MATTER

Important formulas and concepts directly from the NCERT-

- the cathode rays consists of streams of fast moving negatively charged particles.
- The value of e/m is independent of nature of material used as cathode or the gas introduced in the discharge tube.

❖ Electron emission-

- The minimum energy required by an electron to escape from the metal surface is called '**work function**' of metal. It is denoted by ϕ_0 . it is measured in eV.
- **$1\text{eV} = 1.6 \times 10^{-19}$ Joule.**
- Work function depends on the properties of metal and the nature of its surface.
- Work function lowest for caesium and highest for platinum.
- The minimum energy required for electron emission can be supplied by three processes-
 - 1) **Thermionic emission** – by suitably heating, sufficient energy can be imparted.
 - 2) **Field emission** – by applying a very strong electric field, electrons can be emitted.
 - 3) **Photoelectric emission** – when light of suitable frequency illuminates a metal surface, electrons are emitted from the metal surface. These light(photo) generated electrons are known as 'photoelectrons'.

❖ Photoelectric effect-

- No electrons were emitted at all if the frequency of the incident light is smaller than a particular frequency, known as '**threshold frequency**'.
- **Threshold frequency depends on the nature of material of emitter plate.**

❖ Effect of intensity of light on photocurrent-

- **Photocurrent increases linearly with intensity of incident light(for a particular frequency).**

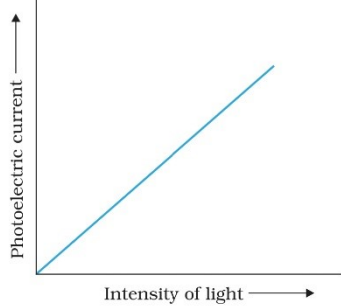


FIGURE 11.2 Variation of Photoelectric current with intensity of light.

- *Photoelectrons emitted per second is directly proportional to intensity.*

❖ Effect of Potential on photocurrent-

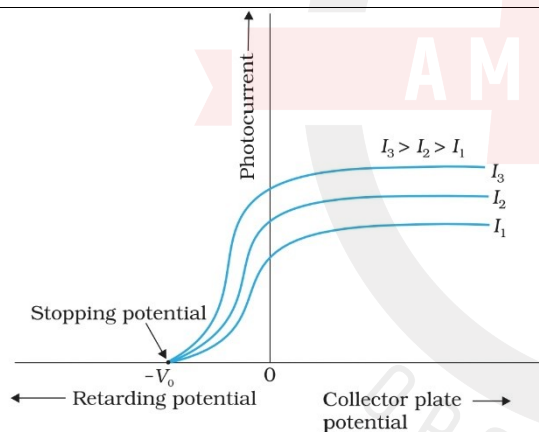


FIGURE 11.3 Variation of photocurrent with collector plate potential for different intensity of incident radiation.

- **For a particular frequency, the minimum negative (retarding) potential V_0 for which photocurrent stops or becomes zero is called ‘cut-off or stopping potential’.**
- Photoelectric current is zero when the stopping potential is sufficient to repel even the most energetic photoelectrons, with the maximum kinetic energy ($K_{\max}$), so that

$$K_{\max} = eV_0$$
- *For a given frequency of incident radiation, the stopping potential is independent of its intensity.*

❖ Effect of frequency on stopping potential-

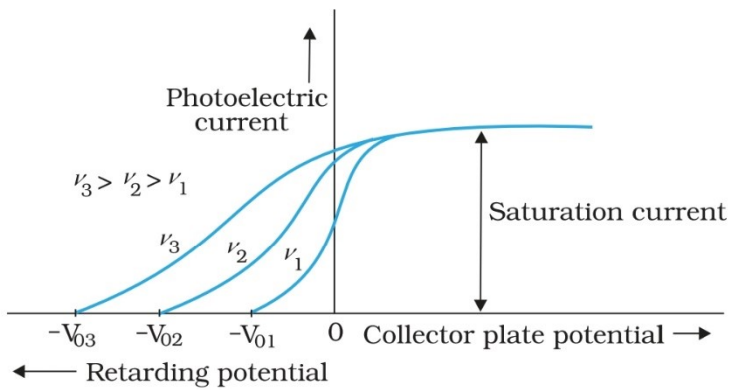


FIGURE 11.4 Variation of photoelectric current with collector plate potential for different frequencies of incident radiation.

- The energy of emitted electrons depends on the frequency of incident radiations.
- **The stopping potential is more negative for higher frequencies of incident radiation.**
- **Greater the frequency of light, greater is the maximum kinetic energy of photoelectrons and greater stopping potential.**

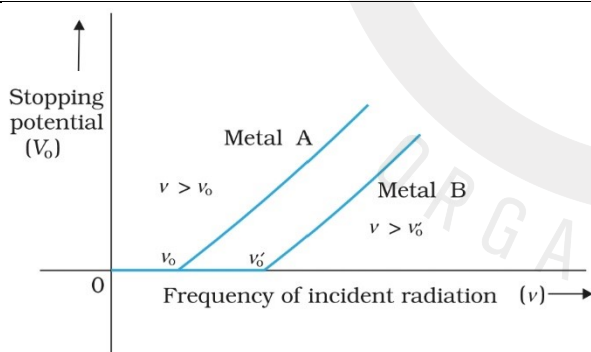


FIGURE 11.5 Variation of stopping potential V_0 with frequency ν of incident radiation for a given photosensitive material.

- This graph shows that-
 - 1) V_0 varies linearly with frequency.
 - 2) There exists a certain minimum cut-off frequency ν_0 for which the stopping potential is zero, known as ‘threshold frequency’
 It is different for different materials.
- $K_{\max}$ is independent of intensity.

- If frequency is lower than threshold frequency, no photoelectric emission takes place.
- Above threshold frequency, the stopping potential or the maximum kinetic energy increases linearly with the frequency of incident radiation, but independent of intensity.
- The photoelectric emission is an instantaneous process.

❖ Einstein's photoelectric equation-

- Radiation energy is built up of discrete units called *quanta of energy of radiation*.
- Each quanta has energy $h\nu$.
- Einstein's photoelectric eqⁿ,

$$K_{\max} = h\nu - \phi_0$$

- Photoelectric emission is possible only if,

$$h\nu > \phi_0$$

$$\text{or } \nu > \nu_0$$

$$\text{where, } \nu_0 = \phi_0/h$$

- $eV_0 = h\nu - \phi_0$
or $V_0 = (h/e)\nu - \phi_0/e$

it predicts that V_0 versus ν curve is a straight line with slope = (h/e) , independent of nature of material.

❖ Particle nature of light-

- Particle nature is denoted by energy packets known as quanta.

- Properties of quanta are-

1) Radiation made up of particles known as photons.

2) Each photon has energy $h\nu$ and momentum, $p = h\nu = h/\lambda$.

3) All photons of a particular frequency has same energy and momentum.

4) Photons are electrically neutral and not deflected by any fields.

5) In a photon-particle collision, total energy and total momentum are conserved.

❖ Wave nature of light-

➤ De Broglie wavelength can be given by,

$$\lambda = h/P = h/mv$$

SUMMARY

1. The minimum energy needed by an electron to come out from a metal surface is called the work function of the metal. Energy (greater than the work function (ϕ_0) required for electron emission from the metal surface can be supplied by suitably heating or applying strong electric field or irradiating it by light of suitable frequency.
2. Photoelectric effect is the phenomenon of emission of electrons by metals when illuminated by light of suitable frequency. Certain metals respond to ultraviolet light while others are sensitive even to the visible light. Photoelectric effect involves conversion of light energy into electrical energy. It follows the law of conservation of energy. The photoelectric emission is an instantaneous process and possesses certain special features.

3. Photoelectric current depends on (i) the intensity of incident light, (ii) the potential difference applied between the two electrodes, and (iii) the nature of the emitter material.
4. The stopping potential (V_0) depends on (i) the frequency of incident light, and (ii) the nature of the emitter material. For a given frequency of incident light, it is independent of its intensity. The stopping potential is directly related to the maximum kinetic energy of electrons emitted:
$$e V_0 = (1/2) m v_{max}^2 = K_{max}$$
5. Below a certain frequency (threshold frequency) ν_0 , characteristic of the metal, no photoelectric emission takes place, no matter how large the intensity may be.
6. The classical wave theory could not explain the main features of photoelectric effect. Its picture of continuous absorption of energy from radiation could not explain the independence of K_{max} on intensity, the existence of ν_0 and the instantaneous nature of the process. Einstein explained these features on the basis of photon picture of light. According to this, light is composed of discrete packets of energy called quanta or photons. Each photon carries an energy $E (= h \nu)$ and momentum $p (= h/\lambda)$, which depend on the frequency (ν) of incident light and not on its intensity. Photoelectric emission from the metal surface occurs due to absorption of a photon by an electron.

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7. Einstein's photoelectric equation is in accordance with the energy conservation law as applied to the photon absorption by an electron in the metal. The maximum kinetic energy $(1/2) m v_{max}^2$ is equal to the photon energy $(h\nu)$ minus the work function $\phi_0 (= h\nu_0)$ of the target metal:

$$\frac{1}{2} m v_{max}^2 = V_0 e = h\nu - \phi_0 = h(\nu - \nu_0)$$

This photoelectric equation explains all the features of the photoelectric effect. Millikan's first precise measurements confirmed the Einstein's photoelectric equation and obtained an accurate value of Planck's constant h . This led to the acceptance of particle or photon description (nature) of electromagnetic radiation, introduced by Einstein.

8. Radiation has dual nature: wave and particle. The nature of experiment determines whether a wave or particle description is best suited for understanding the experimental result. Reasoning that radiation and matter should be symmetrical in nature, Louis Victor de Broglie attributed a wave-like character to matter (material particles). The waves associated with the moving material particles are called matter waves or de Broglie waves.
9. The de Broglie wavelength (λ) associated with a moving particle is related to its momentum p as: $\lambda = h/p$. The dualism of matter is inherent in the de Broglie relation which contains a wave concept (λ) and a particle concept (p). The de Broglie wavelength is independent of the charge and nature of the material particle. It is significantly measurable (of the order of the atomic-planes spacing in crystals) only in case of sub-atomic particles like electrons, protons, etc. (due to smallness of their masses and hence, momenta). However, it is indeed very small, quite beyond measurement, in case of macroscopic objects, commonly encountered in everyday life.

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POINTS TO PONDER

1. Free electrons in a metal are free in the sense that they move inside the metal in a constant potential (This is only an approximation). They are not free to move out of the metal. They need additional energy to get out of the metal.
2. Free electrons in a metal do not all have the same energy. Like molecules in a gas jar, the electrons have a certain energy distribution at a given temperature. This distribution is different from the usual Maxwell's distribution that you have learnt in the study of kinetic theory of gases. You will learn about it in later courses, but the difference has to do with the fact that electrons obey Pauli's exclusion principle.
3. Because of the energy distribution of free electrons in a metal, the energy required by an electron to come out of the metal is different for different electrons. Electrons with higher energy require less additional energy to come out of the metal than those with lower energies. Work function is the least energy required by an electron to come out of the metal.
4. Observations on photoelectric effect imply that in the event of matter-light interaction, *absorption of energy takes place in discrete units of $h\nu$* . This is not quite the same as saying that light consists of particles, each of energy $h\nu$.
5. Observations on the stopping potential (its independence of intensity and dependence on frequency) are the crucial discriminator between the wave-picture and photon-picture of photoelectric effect.
6. The wavelength of a matter wave given by $\lambda = \frac{h}{p}$ has physical significance; its phase velocity v_p has no physical significance. However, the group velocity of the matter wave is physically meaningful and equals the velocity of the particle.

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